

Final Testing & Release Document

This document consolidates every test artefact for the **MidnightZK Off-Ramp SDK** at the **v1.0.0** public release, plus the links to the released SDK and the hosted documentation site, so a reviewer can verify "the SDK is operational without any major errors" from one page.

Item	Status
Tagged release	v1.0.0 — GitHub Release
Hosted developer docs	https://nucastio.github.io/MidnightZK-Off-Ramp-SDK-ADA-Web2-Payments/
Public GitHub repository	https://github.com/Nucastio/MidnightZK-Off-Ramp-SDK-ADA-Web2-Payments
Recorded demo walkthrough	docs/media/offramp-demo.mp4

1. Internal testing (fresh re-run, 2026-06-13)

A fresh run of the internal harness (`npm run test:internal`) was executed against the v1.0.0 build immediately before tagging the release. Result: **29 / 30 successes (96.7%)**, comfortably above the $\geq 90\%$ acceptance threshold. The single failure is an **intentional negative-path** (the mock-mode adapters inject a ~4–6% adapter-rejection rate to exercise the `RailError` path in `submitPayment`).

Run timestamp: 2026-06-13T10:35:32.919Z **Total simulated off-ramps:** 30 (10 × cashapp / wise / revolut)
Overall success rate: 96.7% — PASS (threshold $\geq 90\%$) **Avg proof-generation latency:** 744.6 ms — PASS (NFR-2 budget $\leq 50\,000$ ms) **Avg proof-verification latency:** 0.21 ms

Rail	Runs	Successes	Failures	Success rate	Avg prove (ms)	Avg verify (ms)	Avg submit (ms)	Avg attest (ms)
cashapp	10	10	0	100.0%	742.69	0.17	0.27	0.85
wise	10	10	0	100.0%	763.34	0.19	1778.12	0.55
revolut	10	9	1	90.0%	727.76	0.28	0.17	0.46

The full report (per-step latency, methodology, issues-and-fixes) is in [internal-testing-report.md](#) .

```
Reproduce: npm run test:internal from a clean checkout. Rewrites docs/internal-testing-report.md
+ data/testing-report.json .
```

2. Cardano Preprod end-to-end

Five real Preprod transactions exercise the **full validator surface** (LOCK / REFUND / RELEASE for both Cash App and Wise paths). All five are confirmed on-chain and linked from [testnet-evidence.md](#) :

Step	Adapter	Tx hash	Cardanoscan
LOCK #1	Cash App	f26f023dfc809cb1adad4830bae0025cbe1334fae9811c7a036239eb85fbc6c3	view
REFUND	sender-signed	a8c50ba93412a26c5401dc4477ea6307ad56c808c99a174ab8a69c7675c6d0b9	view
LOCK #2	Wise	03089ef869daf44c511539c915bc825435c18770071aa923322b43b29dc3b869	view
LOCK #3	Wise	b55e48084290f6b88b8fd6489f40e65acc50664fba4873feb1248dfbcb64ac2	view
RELEASE	operator-signed	c84c242d6f86dbdac54ded62c92bbdc88b5725722d1691728854e20d62bd3168	view

- **Validator:** cardano/escrow/validators/escrow.ak, Plutus V3, Aiken v1.1.21.
- **Script address (Preprod):** `addr_test1wrvzmkxhfm9j0u8g6p4cpkevqja4tn8qr88z7l7nc2tqrsxln25g`
- **Datum shape recorded on-chain:** `intent_id`, `payee_commitment`, `amount_commitment`, `adapter_tag`, `sender_pkh`, `operator_pkh`, `deadline`, `vk_hash`, `principal_lovelace` (each LOCK tx in the table above carries a concrete instance).

3. Wise sandbox — live provider integration

The Wise sandbox adapter was driven end-to-end against the public Wise sandbox (<https://api.sandbox.transferwise.tech>) with `RAIL_ADAPTER_MODE=sandbox`. Six raw provider request/response captures are committed under [docs/sandbox-evidence/](#), demonstrating real off-ramp flows when credentials are configured.

Step	File	API path	Method	HTTP
1. List profiles	01-profiles.json	/v1/profiles	GET	200
2. Create quote (USD → USD, 1.50)	02-quote.json	/v3/profiles/{id}/quotes	POST	200
3. Create recipient	03-recipient.json	/v1/accounts	POST	200
4. Create transfer	04-transfer.json	/v1/transfers	POST	200
5. Fund transfer	05-fund.json	/v3/profiles/{id}/transfers/{tid}/payments	POST	403 (SCA-gated; provider-side)
6. Check transfer status	06-status.json	/v1/transfers/{tid}	GET	200

Real Wise transfer: `2147582543`, sandbox profile `30539072`, quote `fedc7ae5-c015-4e37-bb71-404104419610`, recipient `702406717`. Status `incoming_payment_waiting` (the documented Wise state for an unfunded transfer; funding completion requires either an OAuth-token profile or Wise SCA approval and is provider-side).

Cash App uses **Afterpay sandbox** semantics — canonical provider reference: <https://www.postman.com/afterpay-1-426879/afterpay-online-apis-v2/folder/zohg5nd/checkouts>. Revolut follows the same RailAdapter interface and is ready for credentials.

4. Midnight ZK circuit deployment

- **Compact source:** `contract/src/offramp.compact`
- **Compiled artefacts (committed):** `contract/src/managed/`
- **Predicates:** `provePayeeBinding`, `proveAmountBinding`, `proveComplianceFlag`, `proveOffRampSettlement`
- **Contract address:** `a3a72a55eb37317300a6c0718578a8e82040dacce3f139e23e1f52af99f77030`
- **Deploy tx:** `bb81cf19...6e9f2` (block 15768)
- **4 SNARK proofs** deployed across blocks 15774→15784 (per-block detail in `testnet-evidence.md`).

5. Backend API

`tsx backend/api/main.ts` boots a Hono HTTP server that serves the entire off-ramp lifecycle as REST. The OpenAPI 3.0.3 spec is generated by `backend/api/openapi.ts` and exposed live at `/docs` (Swagger UI) and `/api/openapi.json` (machine-readable). All routes are documented in the [API reference](#).

A live evaluation-window deployment is fronted by a TryCloudflare tunnel — see the [README](#) for the current URL. The tunnel is ephemeral; reproducing the run requires only `npm run dev` per the [quickstart](#).

6. Issues fixed before release

Issue	Fix	Where
<code>libsodium-wrappers-sumo</code> ESM resolution — npm publish layout puts <code>libsodium-sumo.mjs</code> in a sibling package; ESM build expects it next to <code>libsodium-wrappers.mjs</code> .	One-time shell fixup that copies the file into the expected location.	<code>scripts/fix-libsodium.sh</code>
Mock-mode rail rejection wasn't reflected in the harness summary cleanly.	Per-rail success-rate column + dedicated "Failures" section in the regenerated report.	<code>scripts/internal-test.ts</code>
Wise sandbox <code>/payments</code> SCA-gating.	Documented as expected provider-side behaviour; the SDK's responsibility ends at submitting the funding intent.	<code>docs/sandbox-evidence/README.md</code> §"Step 5 note"

7. Acceptance-criteria checklist

Milestone criterion	Result	Evidence
Comprehensive developer docs published (integration, API reference, examples)	✓	Docs site — index / quickstart / integration / api-reference / sdk-reference / examples / architecture
Final testing + release document with all test results, SDK link, docs link	✓	This page
Public GitHub repository with final SDK & docs	✓	https://github.com/Nucastio/MidnightZK-Off-Ramp-SDK-ADA-Web2-Payments
Recorded demo walkthrough video	✓	docs/media/offramp-demo.mp4
Tagged release v1.0.0	✓	Release
Transaction success rate $\geq 90\%$	✓	96.7% (29 / 30) — §1 above
Proof generation $\leq 50\,000$ ms (NFR-2)	✓	744.6 ms avg — §1 above
Smart contracts deploy and function on Cardano testnet	✓	5 Preprod txs — §2 above
Real sandbox provider integration	✓	Wise transfer 2147582543 — §3 above
Midnight ZK circuit deployed	✓	4 SNARK proofs in blocks 15774→15784 — §4 above

8. Released artefacts

- **v1.0.0 GitHub Release** — source archive: <https://github.com/Nucastio/MidnightZK-Off-Ramp-SDK-ADA-Web2-Payments/releases/tag/v1.0.0>
- **MIT LICENSE** — <https://github.com/Nucastio/MidnightZK-Off-Ramp-SDK-ADA-Web2-Payments/blob/main/LICENSE>
- **CHANGELOG** — <https://github.com/Nucastio/MidnightZK-Off-Ramp-SDK-ADA-Web2-Payments/blob/main/CHANGELOG.md>
- **Specifications:** [Project Initiation](#) · [SRS](#) · [TAD v1.1](#) (canonical)

9. Documentation link

Hosted developer documentation: <https://nucastio.github.io/MidnightZK-Off-Ramp-SDK-ADA-Web2-Payments/>

Rebuilt automatically on every change to `docs/` or `mkdocs.yml` via `.github/workflows/docs.yml`. The site covers integration, API reference, SDK reference, runnable examples, architecture, and the test/release evidence — sufficient for an external team to integrate the SDK without reading the source.